

Due: Thursday, August 31st at 9:50 am.

Read Section 0.2 and 3.1 in *Levin*. Pay special attention to the definitions of **converse**, **contrapositive** in Section 0.2, and example 3.1.5, for an example of a truth table with P, Q , and R . Then complete the following problems out the book and the problems below. Note that often the solutions are provided in the book problems. Make sure you understand the problem and your own solution and do not just copy the provided solution. For truth tables, only the final column is given, not the full truth table you are expected to create.

- Chapter 0, Section 2: #1 - 5
- Chapter 3, Section 1: #2 - 6
- And complete the following:
 - (1) Show that an implication and its converse are not equivalent with truth tables.
 - (2) Give a real world example of why an implication and its converse are not equivalent.
 - (3) Show that an implication and its contrapositive are equivalent with truth tables.
 - (4) Give a real world example of why an implication and its contrapositive are equivalent.
 - (5) Create a truth table for $P \wedge (\neg Q \rightarrow (R \vee \neg P))$
 - (6) Create a truth table for $(P \vee (Q \wedge R)) \longleftrightarrow ((P \vee Q) \wedge (P \vee R))$. What do you notice about your result?
 - (7) You stumble upon two trolls. You recall that all trolls are either always-truth-telling knights or always-lying knaves. They tell you:
 - Troll 1: If we are cousins, then we are both knaves.
 - Troll 2: We are cousins or we are both knaves.

Could both trolls be knights?